

Are We Running Out of Antibiotics?

What can historical and current data tell us
about antibacterial drug development?

Lisa O'Houlihan (ID6375996)

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Introduction

Antibiotic resistance is a persistent challenge in healthcare, making the development of new antibacterial treatments increasingly important. However, assessing progress in antibacterial development is not as straightforward as counting newly approved drugs. Individual antibiotics may belong to the same class, while current research also includes approaches that differ from traditional antibiotics. This project investigates what historical and current data can reveal about antibacterial drug development. Two publicly available datasets were analysed separately: historical US Food and Drug Administration (FDA) antibiotic approvals and the World Health Organization (WHO) 2025 antibacterial clinical pipeline. Rather than combining them into a single timeline, the analyses examine different aspects of development, including individual drug approvals, antibiotic-class diversity and current treatment approaches.

Data sample

The historical dataset contains 157 antibiotic entries, with FDA approval years ranging from 1939 to 2017. Each entry includes information about the antibiotic, its class and its approval year. Antibiotic classes represent groups of related drugs, allowing individual approvals to be distinguished from the first recorded appearance of each class. The WHO 2025 dataset contains 90 antibacterial products in clinical development. It provides information about product type (traditional or non-traditional), categories of non-traditional approaches and clinical development phase. However, the datasets differ in geographical scope and development stage. The historical dataset records US regulatory approvals, whereas the WHO dataset provides a global view of products undergoing clinical development. Consequently, the two datasets were not treated as directly comparable measurements. The datasets used in this analysis were obtained from publicly available historical antibiotic approval data and the WHO 2025 antibacterial development report [1, 4].

Data Analysis Methods

Both Excel datasets were analysed using Python, primarily with the pandas and Matplotlib libraries. For the historical dataset, approval years were converted into decades using integer division so that individual approvals could be counted per decade. The data were then grouped by antibiotic class, and the earliest FDA approval year recorded for each class was identified. These years were grouped by decade to determine when classes first appeared in the dataset. For the WHO dataset, products were grouped into traditional and non-traditional categories. The frequencies of different non-traditional approaches were calculated, and product types were cross-tabulated against clinical development phases. Inconsistent phase labels, such as NDA/MAA and NDA / MAA, were standardised before counting. Since the dataset contained different numbers of traditional and non-traditional products, development-phase counts were converted into percentages within each product type. This allowed their distributions to be compared without relying solely on absolute counts. Missing category information was also checked to avoid treating incomplete records as fully classified products.

Results and discussion

Individual antibiotic approvals peaked during the 1980s, with 31 approvals. However, when the earliest FDA approval recorded for each antibiotic class was considered, first class appearances peaked in the 1950s, with nine classes. Of the 90 products in the WHO clinical pipeline, 50 were traditional (55.6%) and 40 were non-traditional (44.4%). Among the 40 non-traditional products, 38 had a specific approach recorded. The largest categories were bacteriophages and phage-derived enzymes (14 products), antibodies (10 products) and microbiome-modulating agents (7 products). Development-phase analysis also revealed differences between product types. Combining Phase 3 and NDA/MAA as later development stages, 22% of traditional products (11 of 50) were in these stages, compared with 10% of non-traditional products (4 of 40). These results show that non-traditional approaches represent a substantial proportion of the current portfolio, although a smaller proportion of these products have reached later development stages.

Figure 1
US Antibiotic Approvals by Decade

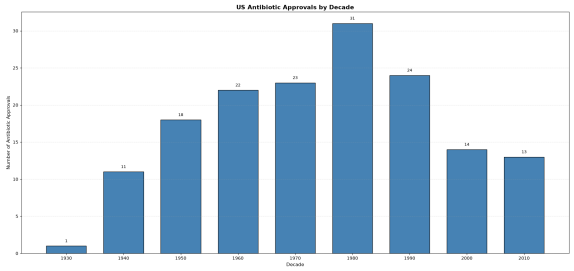
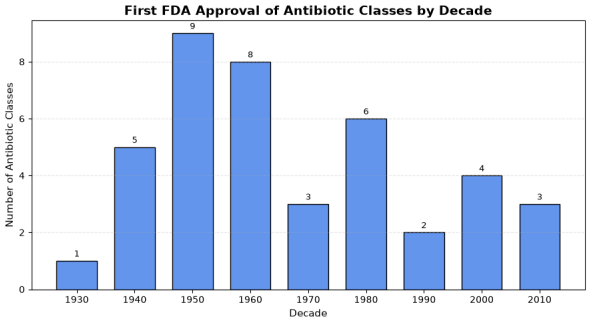


Figure 2
First FDA Approval of Antibiotic Classes by Decade



Note. Both figures were generated using Python from the historical US FDA antibiotic approvals dataset (1939–2017). Figure 1 shows individual approvals, while Figure 2 shows the first recorded approval of each antibiotic class.

Limitations

The historical and WHO datasets cannot establish a continuous trend in global antibacterial development. They differ in geographical coverage, time period and what they measure: approved drugs versus products undergoing clinical development. Furthermore, the earliest approval recorded for a class represents its first appearance within the historical dataset, not necessarily the worldwide discovery or introduction of that class. Similarly, inclusion in the WHO clinical pipeline does not guarantee eventual regulatory approval. A more direct historical-to-current comparison would require datasets with consistent geographical coverage, definitions and measurements across the full period.

Conclusion

This project demonstrates how different methods of organising and analysing data can influence our understanding of antibacterial development. Historical US antibiotic approvals peaked in the 1980s, while the first recorded appearances of antibiotic classes peaked 30 years earlier. This shows why counting individual drugs alone does not fully describe changes in antibiotic-class diversity. Separately, the WHO 2025 pipeline shows that 44.4% of products are non-traditional, reflecting the variety of antibacterial approaches currently under development. Although the datasets cannot establish whether antibacterial development is increasing or decreasing over time, they provide complementary perspectives on the broader research question. By applying data cleaning, grouping, filtering, cross-tabulation and percentage calculations, the analysis goes beyond simple drug counts to characterise different aspects of antibacterial development.

References

[1] Dattani, S. (2024, December 23). *What was the Golden Age of Antibiotics, and how can we spark a new one?* Our World in Data. `https://ourworldindata.org/golden-age-antibiotics`

[2] Hutchings, M. I., Truman, A. W., and Wilkinson, B. (2019). Antibiotics: Past, present and future. *Current Opinion in Microbiology, 51*, 72–80. `https://doi.org/10.1016/j.mib.2019.10.008`

[3] World Health Organization. (2025). *Antibacterial products in clinical development for bacterial priority pathogens.* `https://www.who.int/observatories/global-observatory-on-health-research-and-development/monitoring/antibacterial-products-in-clinical-development-for-priority-pathogens/`

[4] World Health Organization. (2025). *Analysis of antibacterial agents in clinical and preclinical development: Overview and analysis 2025.* `https://www.who.int/publications/i/item/9789240113091`